

Exclusion List ETL —Issues, Resolutions & Open Decisions

医疗 Exclusion List ETL —问题、解决方案与待决策事项

Project / 项目: Six-State Medical Exclusion List Merge (MD, MA, MI, MS, MT, NE)
六州医疗 Exclusion List 合并 (MD, MA, MI, MS, MT, NE)

Status / 状态: Data cleaning complete; awaiting leadership decisions before national merge
数据清洗已完成 ; 全国合并前待领导决策

Cleaned records / 清洗后记录总数: 8,575

Part 1 —Issues Identified and Resolved

第一部分—已识别并解决的问题

1. Michigan: Incorrect Dual Sanction Date Mapping

1. Michigan (密歇根州): 双制裁日期映射错误

EN	Michigan source rows contain two sanction date columns (Sanction Date1, Sanction Date2). Initially, Sanction Date2 was mapped to REINDATE (reinstatement date). This was semantically incorrect —Sanction Date2 represents a second sanction event, not a reinstatement.
中文	Michigan 源数据包含两个制裁日期列 (Sanction Date1、Sanction Date2)。最初将 Sanction Date2 映射到了 REINDATE (恢复纳入医保日期)。这在业务语义上是错误的——Sanction Date2 代表的是第二次制裁事件，而非恢复日期。
EN —Resolution	Each parsed sanction date now generates a separate exclusion record with its own EXCLDATE and EXCLTYPE. REINDATE is left empty. A dedicated parser (michigan_dates.py) handles compound date strings (e.g. 2/14/18 6/14/18). Result: 3,982 source rows → 4,921 cleaned records.
中文—解决方案	每个解析出的制裁日期现在生成一条独立的 exclusion 记录，各有自己的 EXCLDATE 和 EXCLTYPE，REINDATE 置空。并新增专用解析模块 (michigan_dates.py) 处理复合日期字符串 (如 2/14/18 6/14/18)。结果：3,982 行源数据 → 4,921 条清洗记录。

2. Mississippi: Indefinite Exclusion End Date Should Be Empty

2. Mississippi (密西西比州) : 无限期 exclusion 结束日期应置空

EN	Mississippi uses Termination End Date = 2999-12-31 (Excel serial 401768) and Exclusion Period ending in "Indefinite" to indicate exclusions with no fixed end. These were incorrectly converted to REINDATE = 29991231, implying a reinstatement date in year 2999.
中文	Mississippi 使用 Termination End Date = 2999-12-31 (Excel 序列号 401768) 及以 "Indefinite"结尾的 Exclusion Period 来表示无固定结束日的排除。此前被错误转换为 REINDATE = 29991231 , 相当于在 2999 年有一个恢复日期。

EN —Resolution	Records with indefinite exclusion period → REINDATE set to empty. Only records with a real bounded end (e.g. October 30, 2002 – November 19, 2028) retain a populated REINDATE (1 record). Previously affected: 192 records; now: 0 incorrect REINDATE values.
中文—解决方案	无限期排除的记录 → REINDATE 置空。仅有明确结束期的记录 (如 October 30, 2002 – November 19, 2028) 保留 REINDATE (共 1 条)。此前受影响 192 条 ; 现已修正为 0 条错误的 REINDATE。

Part 2 —Open Issues Requiring Leadership Decision

第二部分—待领导决策的遗留问题

Issue 1: Full Names Not Truncated —PostgreSQL Fields Changed to TEXT

问题 1 : 全名不截断—PostgreSQL 字段改为 TEXT

EN —Current approach	Per business requirement, organization names and provider names are preserved in full. PostgreSQL columns (busname, general, lastname, etc.) use TEXT instead of OIG LEIE fixed-length limits (e.g. BUSNAME max 30 chars, GENERAL max 20 chars).
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中文—当前做法	按业务要求，机构名和 provider 姓名完整保留，不做截断。PostgreSQL 相应列 (busname、general、lastname 等) 使用 TEXT 类型，而非 OIG LEIE 固定长度限制 (如 BUSNAME 最长 30 字符、GENERAL 最长 20 字符)。
EN —Scale	272 records have busname exceeding 30 characters. Longest observed: 115 characters. Example: INFINITY INTEGRATED COUNSELING & HOLISTIC WELLNESS (50 chars).
中文—规模	共 272 条记录的 busname 超过 30 字符，最长 115 字符。示例：INFINITY INTEGRATED COUNSELING & HOLISTIC WELLNESS (50 字符)。
EN —Potential risk	If the national exclusion_main table or downstream systems enforce strict OIG fixed-length fields, import may fail or values may be silently truncated. Data will not match the canonical CMS OIG LEIE file format.
中文—潜在风险	若全国 exclusion_main 总表或下游系统强制要求 OIG 固定长度字段，导入可能失败或被静默截断，与 CMS OIG LEIE 标准文件格式不完全一致。
EN —Options for leadership	(A) National table uses TEXT —no truncation at merge (recommended). (B) Keep full names in cleaned CSV; truncate only at merge into OIG-compliant main table, with a truncation log. (C) Confirm field definitions with Verbus before merge.
中文—待领导选择	(A) 全国总表统一使用 TEXT，合并时不截断 (推荐)。 (B) 清洗库保留全名，仅在合并入 OIG 合规总表时截断并记录日志。 (C) 与 Verbus 确认全国表字段定义后再决定。

Issue 2: Source Records Missing Effective Date (8 Records Total)

问题 2：源数据缺少生效日期 (共 8 条)

2a. Montana (MT) —6 records

EN	Six a.k.a. (alias) provider records in the Montana source file have no Effective Date. Names are now parsed correctly (e.g. a.k.a. Saul Clement → Last: Clement, First: Saul), but EXCLDATE cannot be populated from source.
中文	Montana 源文件中有 6 条 a.k.a. (别名) provider 记录无 Effective Date。姓名现已正确解析 (如 a.k.a. Saul Clement → 姓 Clement , 名 Saul), 但 EXCLDATE 无法从源数据填充。
EN —Impact	These records cannot be matched by exclusion date in claims screening. Matching by provider name remains possible. This is a source data gap, not a conversion error.
中文—影响	这些记录无法通过排除日期在理赔筛查中命中, 但仍可通过 provider 姓名匹配。属于源数据缺失, 非转换错误。
2b. Mississippi (MS) —2 records	
EN	Two Seyah Hospice Care organization records in the Mississippi source Excel have a blank Termination Effective Date (NaT). A third related record for the same organization with date 2012-02-01 is correctly preserved.
中文	Mississippi 源 Excel 中两条 Seyah Hospice Care 机构记录的 Termination Effective Date 为空 (NaT)。同机构另一条含日期 2012-02-01 的记录已正确保留。
EN —Impact	Same as MT —date-based screening unavailable for these 2 records; name/NPI matching still possible. Source data limitation.
中文—影响	与 MT 情况相同——这 2 条无法按日期筛查, 仍可通过姓名/NPI 匹配。属源数据限制。
EN —Options for leadership	(A) Retain records and add a data_quality_flag = missing_excldate field. (B) Exclude from main table until manually verified. (C) Contact state data source for correction.

中文—待领导选择	(A) 保留记录并增加 data_quality_flag = missing_excldate 标记。(B) 暂不导入总表，待人工核实。(C) 联系各州数据源请求补全。
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Issue 3: Sparse Fields in Source Data (Structural Limitation)

问题 3：源数据字段稀疏（结构性限制）

State / 州	Missing field / 缺失字段	Extent / 比例	Cause / 原因
Nebraska (NE)	NPI	82% (1,134 / 1,390)	PDF table does not include NPI for most providers
Nebraska (NE)	Address	100% (1,390 / 1,390)	PDF table has no address column
Massachusetts (MA)	Address	100% (294 / 294)	Source Excel has no address column

EN	These are inherent limitations of the state-published source files, not conversion errors. Claims matching for these states will rely primarily on NPI and/or provider name.
中文	这些是州政府发布的源文件本身的局限，非转换错误。这些州的理赔匹配将主要依赖 NPI 和/或 provider 姓名。

Issue 4: Cross-State Duplicate NPI —Retain Multiple Records or Deduplicate?

问题 4：跨州相同 NPI —保留多条还是去重？

EN —Current state	15 NPIs appear in multiple source_state values. Example: NPI 1538334222 (Adam Medical Equipment) appears in MD, MI, and NE —likely the same organization listed independently by each state's Medicaid program.
中文—现状	15 个 NPI 同时出现在多个 source_state 中。示例：NPI 1538334222 (Adam Medical Equipment) 同时出现在 MD、MI、NE 三个州——很可能是同一机构被各州 Medicaid 项目独立收录。

EN —Options	(A) Retain all records —each row keeps its <code>source_state</code> ; supports audit trail and state-specific provenance (recommended). (B) Cross-state deduplication —keep one record per NPI in the national table; simpler but loses state-level sourcing.
中文—选项	(A) 保留全部记录—每条保留其 <code>source_state</code> ，便于审计溯源（推荐）。(B) 跨州去重—全国总表每个 NPI 只保留一条；更简洁但丢失州级来源信息。
EN —Recommendation	Retain multiple records at merge; use <code>source_state</code> to distinguish origin. Align rule with Verbus before inserting into national <code>exclusion_main</code> .
中文—建议	合并时保留多条，以 <code>source_state</code> 区分来源；插入全国 <code>exclusion_main</code> 前与 Verbus 统一规则。

Part 3 —Summary for Leadership

第三部分—领导汇报摘要

Category / 类别	EN	中文
Resolved	Michigan dual-date mapping fixed; MS indefinite REINDATE cleared; dedup logic redesigned; MD legend rows filtered; MT alias names parsed; EXCLTYPE exact-match fixed	Michigan 双日期映射已修正；MS 无限期 REINDATE 已置空；去重逻辑已重设计；MD 图例行已过滤；MT 别名姓名已修正；EXCLTYPE 精确匹配已修复
Data quality	6/6 states PASS validation; 0 high-risk conversion errors; 8,575 cleaned records	六州校验全部 PASS；0 项高风险转换错误；共 8,575 条清洗记录
Needs decision	Full-name TEXT vs OIG truncation; 8 missing-date records; cross-state NPI policy; national table schema with Verbus	全名 TEXT vs OIG 截断；8 条无日期记录处理；跨州 NPI 策略；与 Verbus 确认全国表结构
Not yet done	PostgreSQL import on server; merge into 50-state <code>exclusion_main</code>	PostgreSQL 服务器导入；合并入 50 州 <code>exclusion_main</code> 总表

Appendix —Record Counts by State

附录—各州记录数

State	Source rows / 源文件行数	Cleaned records / 清洗后记录	Notes / 备注
MD	1,605	1,603	6 legend rows filtered / 过滤 6 条图例行
MA	294	294	No address in source / 源文件无地址
MI	3,982	4,921	Multi-date expansion / 多日期展开
MS	194	193	2 missing effective date in source / 源文件 2 条无生效
MT	174	174	6 alias records missing date in source / 6 条别名记录
NE	1,391	1,390	82% missing NPI; no address / 82% 无 NPI ; 无地址
Total	7,640	8,575	

Document generated for team reporting. 本文档供团队汇报留存使用。